

Type of array

Difficulty: Basic | Accuracy: 26.04% | Submissions: 80K+ | Points: 4

You are given an array `arr[]` having unique elements. Your task is to return the type of array described below.

- Return 1 if the array is in ascending order.
- Return 2 if the array is in descending order.
- Return 3 if the array is in descending rotated order.
- Return 4 if the array is in ascending rotated order.

You may assume that the input array is always one of the four types.

Examples

Input: `arr[] = [2, 1, 5, 4, 3]`

Output: 3

Explanation: Descending rotated, rotate 2 times left.

Input: `arr[] = [3, 4, 5, 1, 2]`

Output: 4

Explanation: Ascending rotated, rotate 2 times right.

Constraints

$$3 \leq \text{arr.size()} \leq 10^5$$

$$1 \leq \text{arr}[i] \leq 10^6$$

Expected Complexities

Time Complexity: $O(n)$

Auxiliary Space: $O(n)$

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